

DAY 4 ... TASK 4:

INHERITANCE & POLYMORPHISM. REUSABILITY AND FLEXIBILITY

Deeply understand and implement **Inheritance & Polymorphism**, both in theory and practice

✓ Assigned Task 4

1. Theoretical Section *(Must Write in Your Own Words After Learning)*

◆ Inheritance Learn & Explain:

1. What is inheritance?
2. Types of inheritance in Python:
 - Single
 - Multiple
 - Multilevel
 - Hierarchical
 - Hybrid (mention limitations in Python)
3. What is the use of `super()` in inheritance?
4. Why inheritance is important in large-scale projects?
5. What is the problem with multiple inheritance? (MRO – Method Resolution Order)

◆ Polymorphism Learn & Explain:

1. What is polymorphism in OOP?
2. Difference between:
 - Compile-time vs Runtime polymorphism (mention Python's dynamic nature)
3. What is method overriding?
4. What is duck typing in Python?
5. How does polymorphism increase flexibility in code?

Write all answers clearly as notes (you'll revise before your viva...).

2. Practical Coding Task

➤ Scenario:

Build a system for a **Zoo Management App** that uses inheritance and polymorphism to manage animals.

❖ Your Requirements:

◆ Create a Base Class: Animal

- Attributes:
 - name, species
- Method:
 - make_sound() – returns "Some generic sound"

◆ Create 3 Child Classes:

1. Dog – Overrides make_sound() → "Bark!"
2. Cat – Overrides make_sound() → "Meow!"
3. Lion – Overrides make_sound() → "Roar!"

◆ Use super() to initialize base class in each child.

◆ Now create a method:

```
def animal_sound(animal: Animal):  
    animal.make_sound()
```

Call this method for all 3 types **showing polymorphism** in action.

➔ Bonus Coding Challenge (Optional):

- Add a new method info() in base class and override it in child classes.
 - Show multiple inheritance by creating a class TrainedDog that inherits from both Dog and another class Training.
-

➔ **Bonus Tips:**

- Override methods properly.
 - Think real-world: reuse parent code using `super()`.
 - Use type hinting and comments.
-

 Deadline:

Submit by: *Tomorrow before 11:59 PM*

Day: 4 of 8

This is your core viva topic master it like a boss 😎

Final :

Once done, reply with:

“ Task 4 Completed “

And I'll move you straight to the next big step to master advanced OOP!....